Problem

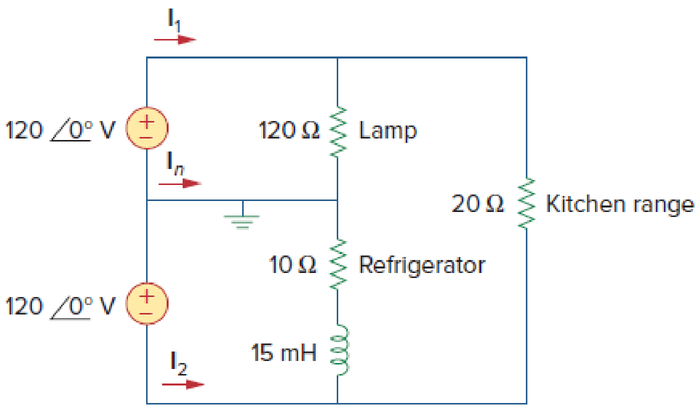
A regular household system of a single-phase three-wire circuit allows the operation of both 120-V and 240-V, 60-Hz appliances. The household circuit is modeled as shown in Fig. 11.96. Calculate:

(a) the currents **I**1, **I**2, and **I***n*

(b) the total complex power supplied

(c) the overall power factor of the circuit

Figure 11.96:



Step-by-step solution

Step 1 of 11

Refer to Figure 11.96 in the text book.

Consider the circuit diagram shown in Figure 1.

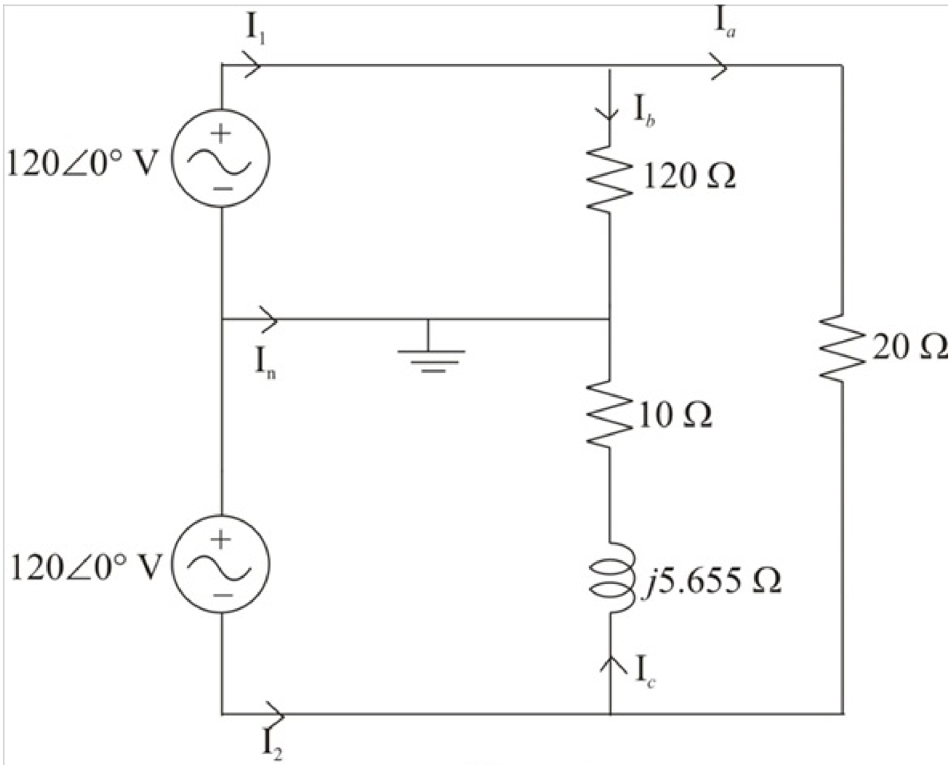


Figure 1

Step 2 of 11

Calculate the voltage across the $20\ \Omega$ resistor.

$$\begin{aligned}V &= 120\angle 0^\circ - (-120\angle 0^\circ) \\ &= 120\angle 0^\circ + 120\angle 0^\circ \\ &= 240\angle 0^\circ\ \text{V}\end{aligned}$$

Therefore, the value of the voltage across the $20\ \Omega$ resistor is $240\angle 0^\circ\ \text{V}$.

Step 3 of 11

Calculate the value of the current I_a.

$$\begin{aligned}I_a &= \frac{240\angle 0^\circ}{20} \\ &= 12\angle 0^\circ\ \text{A}\end{aligned}$$

Therefore, the value of the current I_a is $12\angle 0^\circ\ \text{A}$.

Step 4 of 11

Calculate the value of the current I_b.

$$\begin{aligned}I_b &= \frac{120\angle 0^\circ}{120} \\ &= 1\angle 0^\circ\ \text{A}\end{aligned}$$

Therefore, the value of the current I_b is $1\angle 0^\circ\ \text{A}$.

Step 5 of 11

Calculate the value of the current I_c.

$$\begin{aligned}I_c &= \frac{-120\angle 0^\circ}{10 + j5.655} \\ &= \frac{-120\angle 0^\circ}{11.488\angle 29.488^\circ} \\ &= 10.446\angle 150.512^\circ\ \text{A}\end{aligned}$$

Therefore, the value of the current I_c is $10.446\angle 150.512^\circ\ \text{A}$.

Step 6 of 11

(a)

Calculate the value of the current I_1.

$$I_1 = I_a + I_b$$

Substitute $12\angle 0^\circ\ \text{A}$ for I_a and $1\angle 0^\circ\ \text{A}$ for I_b.

$$\begin{aligned}I_1 &= 1\angle 0^\circ + 12\angle 0^\circ \\ &= 13\angle 0^\circ\ \text{A}\end{aligned}$$

Therefore, the value of the current I_1 is $\boxed{13\angle 0^\circ\ \text{A}}$.

Step 7 of 11

Calculate the value of the current I_2.

$$I_2 = I_c - I_a$$

Substitute $10.446\angle 150.512°\ A$ for I_c and $12\angle 0°\ A$ for I_a.

$$\begin{aligned} I_2 &= 10.446\angle 150.512° - 12\angle 0° \\ &= 21.71\angle 166.3°\ A \end{aligned}$$

Therefore, the value of the current I_2 is $\boxed{21.71\angle 166.3°\ A}$.

Step 8 of 11

Calculate the value of the current I_n.

$$I_n = -(I_b + I_c)$$

Substitute $10.446\angle 150.512°\ A$ for I_c and $1\angle 0°\ A$ for I_b.

$$\begin{aligned} I_n &= -(10.446\angle 150.512° + 1\angle 0°) \\ &= -(9.588\angle 147.57°) \\ &= 9.588\angle -32.43°\ A \end{aligned}$$

Therefore, the value of the current I_n is $\boxed{9.588\angle -32.43°\ A}$.

Step 9 of 11

(b)

Calculate eh value of the total complex power supplied.

$$S_T = V_1(I_1)^* + V_2(I_2)^*$$

Substitute $13\angle 0°\ A$ for I_1, $21.71\angle 166.3°\ A$ for I_2, $120\angle 0°\ V$ for V_1 and $-120\angle 0°\ V$ for V_2.

$$\begin{aligned} S_T &= (120\angle 0°)(13\angle 0°)^* + (-120\angle 0°)(21.71\angle 166.3°)^* \\ &= (120\angle 0°)(13\angle 0°) + (-120\angle 0°)(21.71\angle -166.3°) \\ &= 1560\angle 0° + 2605.2\angle 13.7° \\ &= 4137.35\angle 8.576° \end{aligned}$$

$$S_T = 4091.09 + j616.97\ VA$$

Therefore, the value of the complex power supplied to the load is

$\boxed{4091.09 + j616.97\ VA}$.

Compare $4091.09 + j616.97\ VA$ to $P + jQ$, where P and Q represents the real and reactive power respectively.

Therefore, the values of P and Q are $4091.09\ W$ and $616.97\ VAR$ respectively.

Step 10 of 11

(c)

Calculate the value of the apparent power S.

$$S = \sqrt{P^2 + Q^2}$$

Substitute $4091.09\ \text{W}$ for P and $616.97\ \text{VAR}$ for Q.

$$\begin{aligned} S &= \sqrt{(4091.09)^2 + (616.97)^2} \\ &= \sqrt{16737017.39 + 380651.9809} \\ &= \sqrt{17117669.37} \\ &= 4137.35\ \text{VA} \end{aligned}$$

Therefore, the value of the apparent power is $4137.35\ \text{VA}$.

Step 11 of 11

Calculate the value of the overall power factor.

$$\text{pf} = \frac{P}{S}$$

Substitute $4137.35\ \text{VA}$ for S and $4091.09\ \text{W}$ for P.

$$\begin{aligned} \text{pf} &= \frac{4091.09}{4137.35} \\ &= 0.9888 \end{aligned}$$

Therefore, the value of the overall power factor is $\boxed{0.9888}$.